

6.1 Researching water scarcity

Select ONE of the following case studies of water scarcity. Research the example to answer the questions.

Remember to use valid and reliable sources of information. To ensure you are doing this, check that the information comes from trustworthy organisations, such as government websites, recognised charities, academic sources, or reputable news outlets. Also, cross-check your facts using at least two reliable sources and avoid websites that have no author, no date, or obvious commercial bias.

Select one of the options of water scarcity examples below.

Cape Town, South Africa – “Day Zero” crisis (2018–present)	Bangalore, India – Rapid urbanisation and groundwater stress	Ethiopia – Limited infrastructure and rural water access
Nepal – Water availability vs. lack of safe water supply	California, USA – Droughts and groundwater depletion	Mexico City, Mexico – Sinking city with severe shortages
Cairo, Egypt – Nile dependency and population growth	The Sahel Region (e.g., Niger, Mali)	The Yellow River Basin, China – Overuse and seasonal drought

Part A. Background Information

Provide context of your case study. This is the relevant background information, such as:

- its location (country, region, latitude and longitude)
- describe the climate and population size,
- explain the values of water here and
- include a map

Context:

Part B. Causes of water scarcity

What has caused the water scarcity in this place? Consider the physical (e.g. climate, climate change, geology) and human causes (e.g. population growth, over farming, pollution, poor management). Include statistics and examples.

Answer:

Part C. Impacts of water scarcity

Research and summarise the different impacts of water scarcity on this place and its people.

Environmental (e.g. impacts on lakes, soils, ecosystems, biodiversity)	Social (e.g. impacts on health, communities, sanitation,)	Economic (e.g. impacts on jobs, farming, business, costs)	Political (e.g. conflicts, disagreements, governments)

Part D. Extent and severity of the problem

How widespread and severe is the water scarcity in this place or this example? Consider scale (local, regional, national), how many people are affected and the length of time it occurred or has been occurring. Include one piece of data.

Answer:

Part E. Management strategies and solutions

What has been implemented to overcome water scarcity, and was it effective?

Examples of strategies: Water restrictions, Desalination plants, Building dams/reservoirs, Improving irrigation, Fixing leaks, Reusing/recycling water, Rainwater harvesting or Education campaigns.

Answer:

Part F. Evaluation

What do you think is the most effective solution moving forward for this place to overcome water scarcity, and why?

Answer: